

FULL NAME: _____

BLOCK: _____

Logarithmic Functions + DOK-3 Sales Revenue Inverse (Item 28)

Lesson 6-4 · Quick · Student Packet

OBJECTIVES

Math Objective	Graph logarithmic functions, identify key features (asymptote, domain, range, intercepts), and find equations of inverses of exponential and logarithmic functions.
Language Objective	Describe how transforming a log function shifts its asymptote and key points; explain why the inverse of a log is an exponential.
Essential Understanding	Log and exponential functions are inverses — swap x and y , solve. The asymptote of a log is the y -axis mirror of an exp function's horizontal asymptote.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency with graphing log functions and finding inverses; apply to the Sales Revenue DOK-3 task (Item 28 rehearsal for Form B model-then-invert structure).
Essential Question	How do the graphs of logarithmic and exponential functions relate, and how do we find one from the other?
Materials	Student packet, pencil, calculator. Rules card: $\log_b 1 = 0$; $\log_b b = 1$; $y = \log_b x$ has VA at $x = 0$; $y = \log_b(x - h) + k$ has VA at $x = h$; inverse of $y = a \cdot b^x$ is $y = \log_b(x/a)$; inverse of $y = \log_b(x - h) + k$ is $y = b^{(x-k)} + h$.

LAUNCH — Inverse of a Log Function (teacher models Ex 3B)**[BOOK] LOG FUNCTION KEY FEATURES + INVERSE RULES (keep printed on your page)**LOG FUNCTION KEY FEATURES ($y = \log_b x$, $b > 0$, $b \neq 1$):

- Domain: $x > 0$
- VA: $x = 0$ (y -axis)
- Passes through $(b, 1)$
- Range: all real numbers
- x -intercept: $(1, 0)$

TRANSLATION ($y = \log_b(x - h) + k$):

- VA shifts to $x = h$
- Point $(b + h, 1 + k)$
- Domain: $x > h$

INVERSE RULES:

- $y = a \cdot b^x \Rightarrow$ inverse $y = \log_b(x/a)$
- $y = \log_b(x - h) + k \Rightarrow$ inverse $y = b^{(x-k)} + h$

COMPOSITION CHECK: $f(f^{-1}(x)) = x$ **[WARN] ONE EXAMPLE IN LAUNCH — Ex 3B: inverse of $g(x) = \log_7(x + 5)$** Teacher models Example 3B: find the inverse of $g(x) = \log_7(x + 5)$, then verify using composition $f(f^{-1}(x)) = x$.

Contrast: Ex 3A (inverse of an exponential) covered in 90 sec — same swap-and-solve process, direction

reversed.

Key move: swap x and $y \rightarrow 7^x = y + 5 \rightarrow$ solve for y .

EXPLORE — Think-Pair-Share (25 min)

Work with a partner. Start with [STAR] Practice #28 (DOK-3 Sales Revenue). Then DOK-2 items in order.

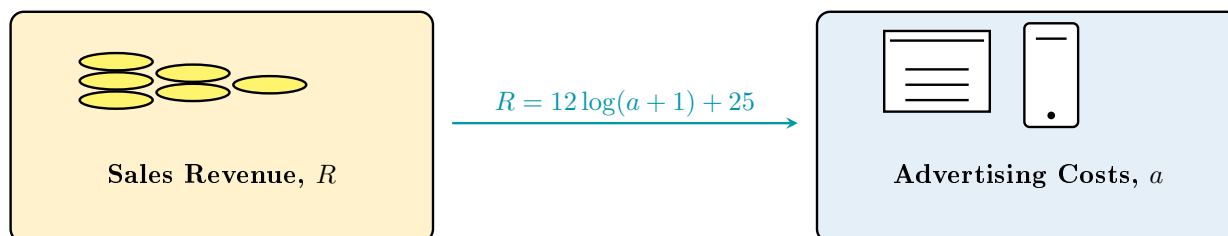
45-min Wed F: q9 is dropped. Skip to Exit after q13.

Bank-item labels (in order):

1. Practice #28 [STAR] DOK-3 SALES REVENUE INVERSE — Form B model-then-invert rehearsal
2. Practice #21 — Inverse of exponential $5^{(x-3)}$ [Form B Q7 rehearsal]
3. Practice #24 — Inverse of $\log_2(8x)$, token drag [Form B Q12 rehearsal]
4. Practice #13 — Asymptote of translated log from graph [Form B Q11 rehearsal]
5. Practice #9 — Are graphs inverses? Inverse from graph [Form B Q9/Q10 rehearsal]

Practice #28 [STAR] DOK-3 SALES REVENUE INVERSE — Form B model-then-invert rehearsal

A company uses this function to relate sales revenue, R , and advertising costs, a . What is the equation of the inverse of the equation? Which equation would be easier to use to find a value of a for a particular value of R ?



Practice #21 — Inverse of exponential $5^{(x-3)}$ [Form B Q7 rehearsal]

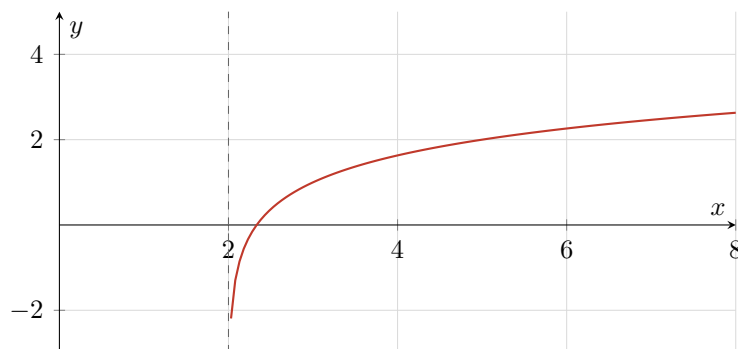
Find the equation of the inverse of each function. $f(x) = 5^{x-3}$.

Practice #24 — Inverse of $\log_2(8x)$, token drag [Form B Q12 rehearsal]

Find the equation of the inverse of each function. $f(x) = \log_2(8x)$.

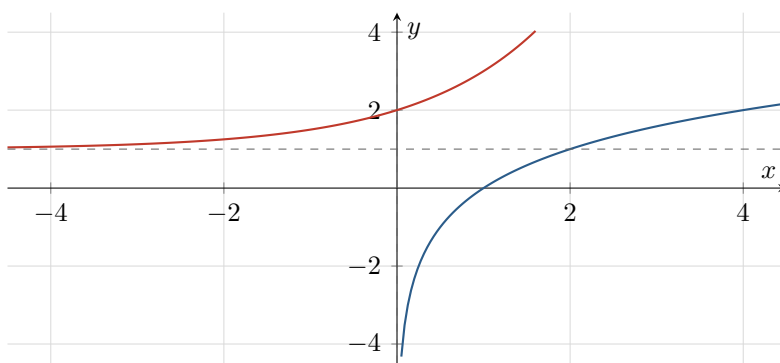
Practice #13 — Asymptote of translated log from graph [Form B Q11 rehearsal]

Use Structure The graph shows a transformation of the parent graph $f(x) = \log_3 x$. Write an equation for the graph.



Practice #9 — Are graphs inverses? Inverse from graph [Form B Q9/Q10 rehearsal]

Look for Relationships Are the logarithmic and exponential functions shown inverses of each other? Explain.



SHARE / SUMMARY (5 min)

Sentence frames

DOK-3 pair share: “For Practice #28, I found the inverse by _____. For revenue $R = \underline{\hspace{1cm}}$ dollars, the original form $R = 12 \cdot \log(a + 1) + 25$ is easier because _____. The inverse form $a = \underline{\hspace{1cm}}$ is easier when _____.”

Bridge to 6-5: log properties let us rewrite and simplify log expressions — that makes inverses and equations easier.

[SEARCH] BRIDGE TO LESSON 6-5

In 6-5 you will learn log properties (product, quotient, power rules). Those properties let you expand or condense log expressions — which makes solving equations like the one in #28 much faster.

EXIT

Find the inverse of $f(x) = 3 \cdot \log_5(x - 2) + 1$.

State the vertical asymptote of f .

State the horizontal asymptote of f^{-1} .

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #33 is a Savvas performance task (telescope) that extends inverse/log fluency to a modeling context.

Optional #1 (Savvas Practice #33 — Telescope PT)

Performance Task The logarithmic function $M(d) = 5 \log d + 2$ is used to find the limiting magnitude of a telescope, where d represents the diameter of the lens of the telescope (mm) that is being used for the observation.

Part A Find the limiting magnitude of a telescope having a lens diameter of 40 mm.

Part B Find the equation of the inverse of this function.

Part C Interpret why astronomers may wish to use the inverse of this function. Justify your reasoning.

Part D Using the inverse function, find the diameter of the lens that has a limiting magnitude of 13.5. Check your answer with the table function of your graphing calculator.